ANGULAR
ARCHITECTS
INSIDE KNOWLEDGE

Angular Initial Load Performance Optimization

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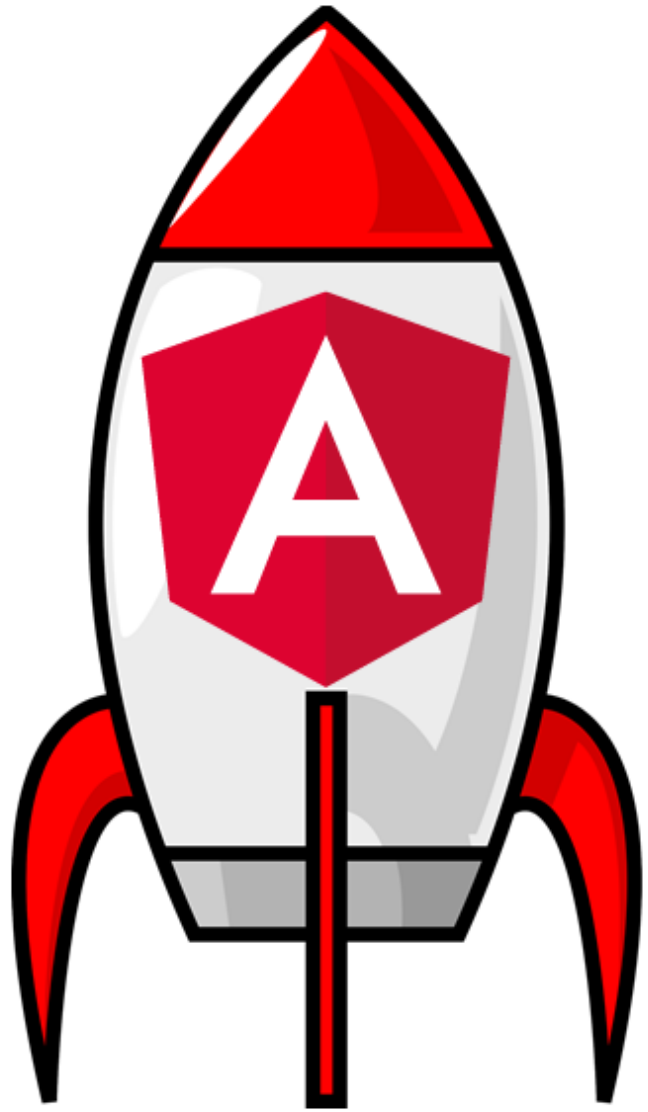
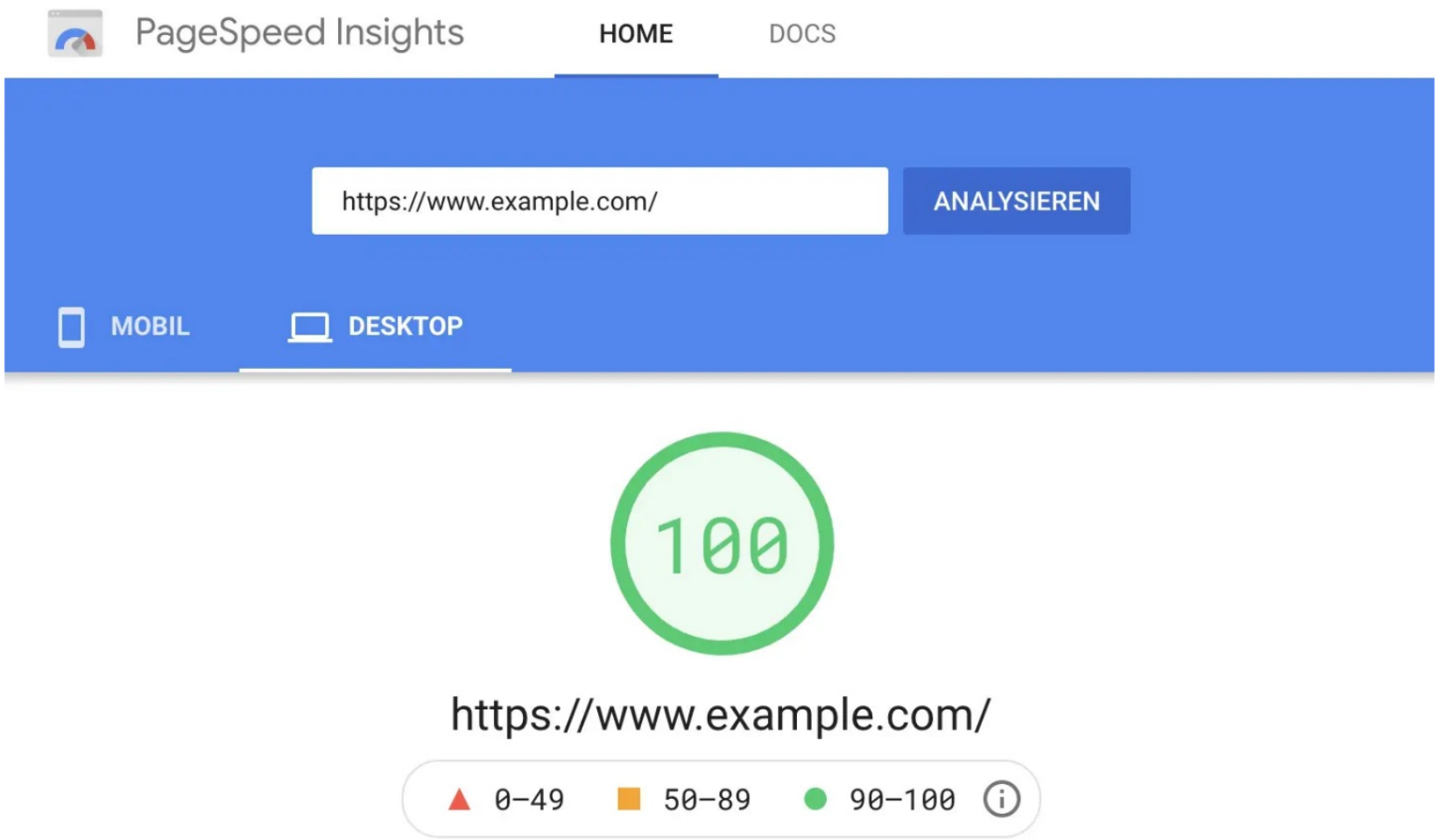


Image from: https://bit.ly/ng-initial-load

Outline

1. Use web performance best practices
2. Use NgOptimizedImage (since NG 14.2.0)
3. Use Build Optimization
4. Avoid large 3rd party libs / CSS frameworks
5. Use Lazy Loading done right
6. Critical Rendering Path / Above the fold
7. Server-side rendering & prerender or cache on the server
8. Use a URL cache

Web Performance Best Practices



#1: Use web performance best practices - I

Problems:

- *Images not optimized*
- *Images not properly sized*
- *Slow server infrastructure*
- *Unused JS code or CSS styles*
- *Too large assets, too many assets*
- *Caching not configured correctly*
- *Compression not configured correctly*
- *…*

#1: Use web performance best practices - II

Identify:

- Lighthouse & PageSpeed Insights
- WebPageTest.org or
- Chrome DevTools

#1: Use web performance best practices - III

Solutions:

- *Images not optimized → Use .webp, .avif or .svg*
- *Images not properly sized → Use srcsets*
- *Slow server infrastructure → HTTP/2, CDN*
- *Unused JS code or CSS styles → Clean up & lazy load assets*
- *Too large assets, too many assets → Clean up & lazy load assets*
- *Caching not configured correctly → Configure it*
- *Compression not configured correctly → Brotli or Gzip*
- *…*

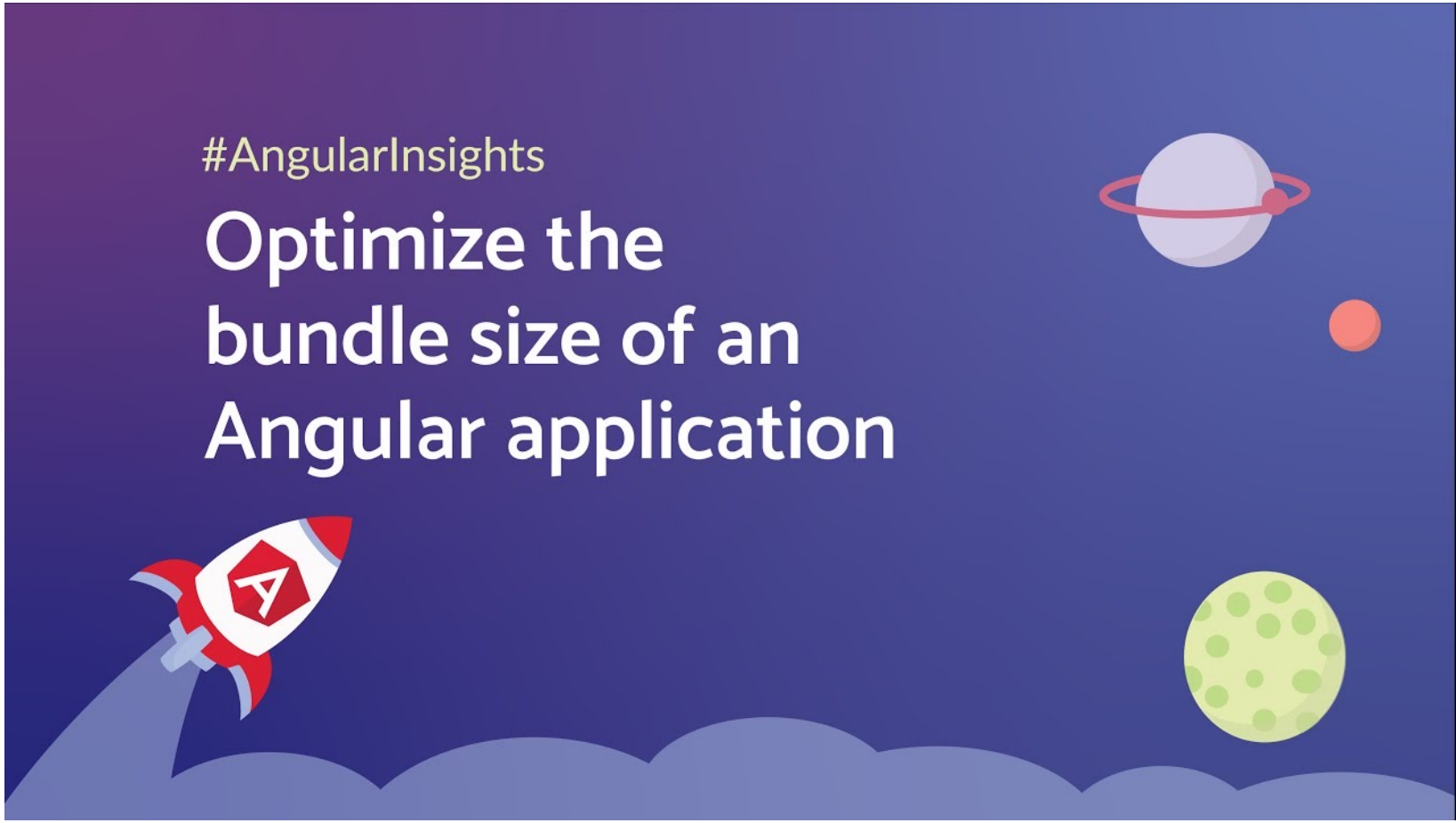
#2: Use NgOptimizedImage (since NG 14.2.0)

- Problem: *Lighthouse or PageSpeed report image errors / warnings*

- Identify: Lighthouse & PageSpeed Insights / WebPageTest or DevTools

- Solution: Use NgOptimizedImage's ngSrc instead of src attribute
  - Takes care of intelligent lazy loading
  - Prioritization of critical images ("above-the-fold")
  - Also creates srcset & sizes attributes (for responsive sizes, since NG 15.0.0)
  - Also supports high-resolution devices ("Retina images")

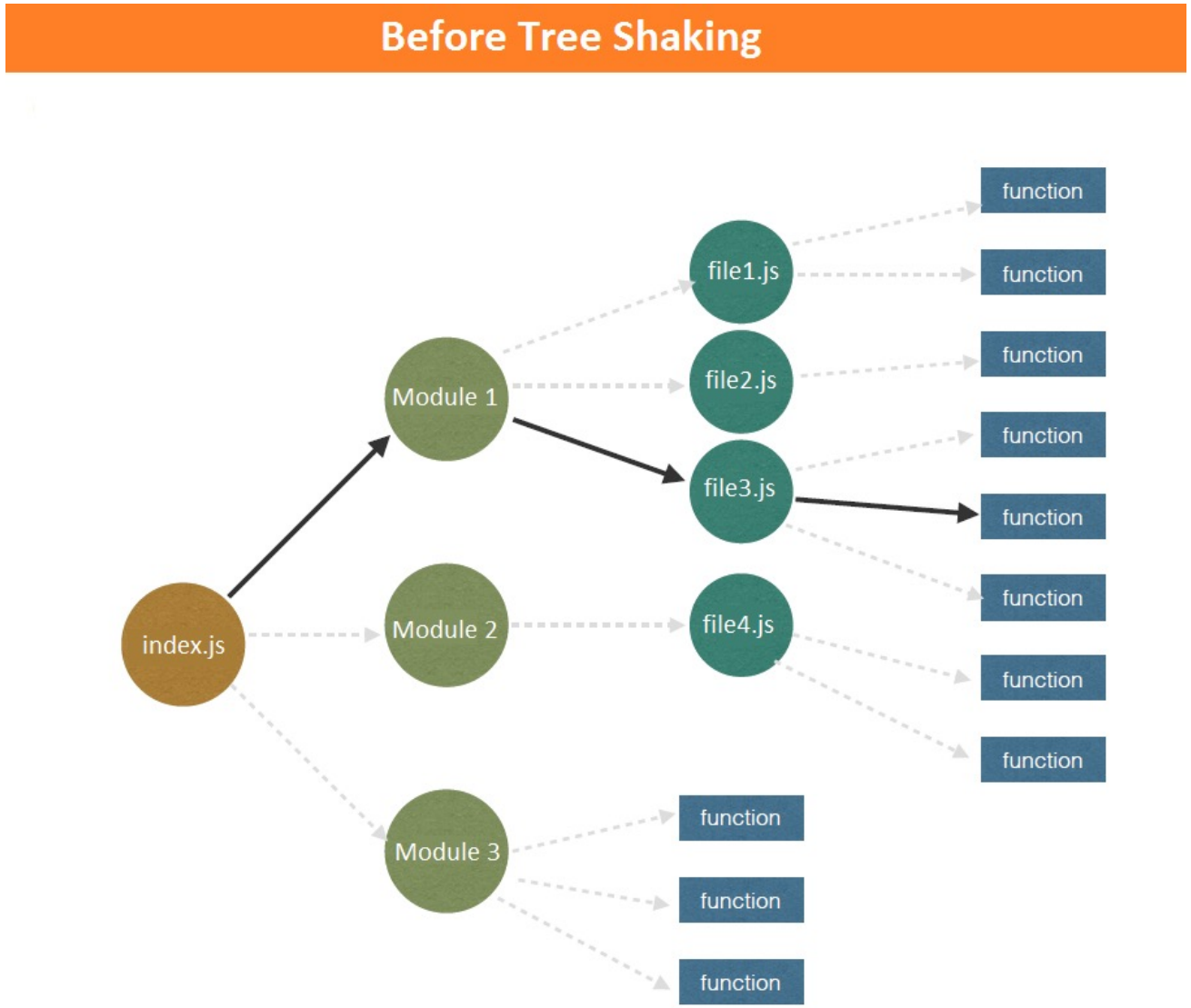
Build optimization



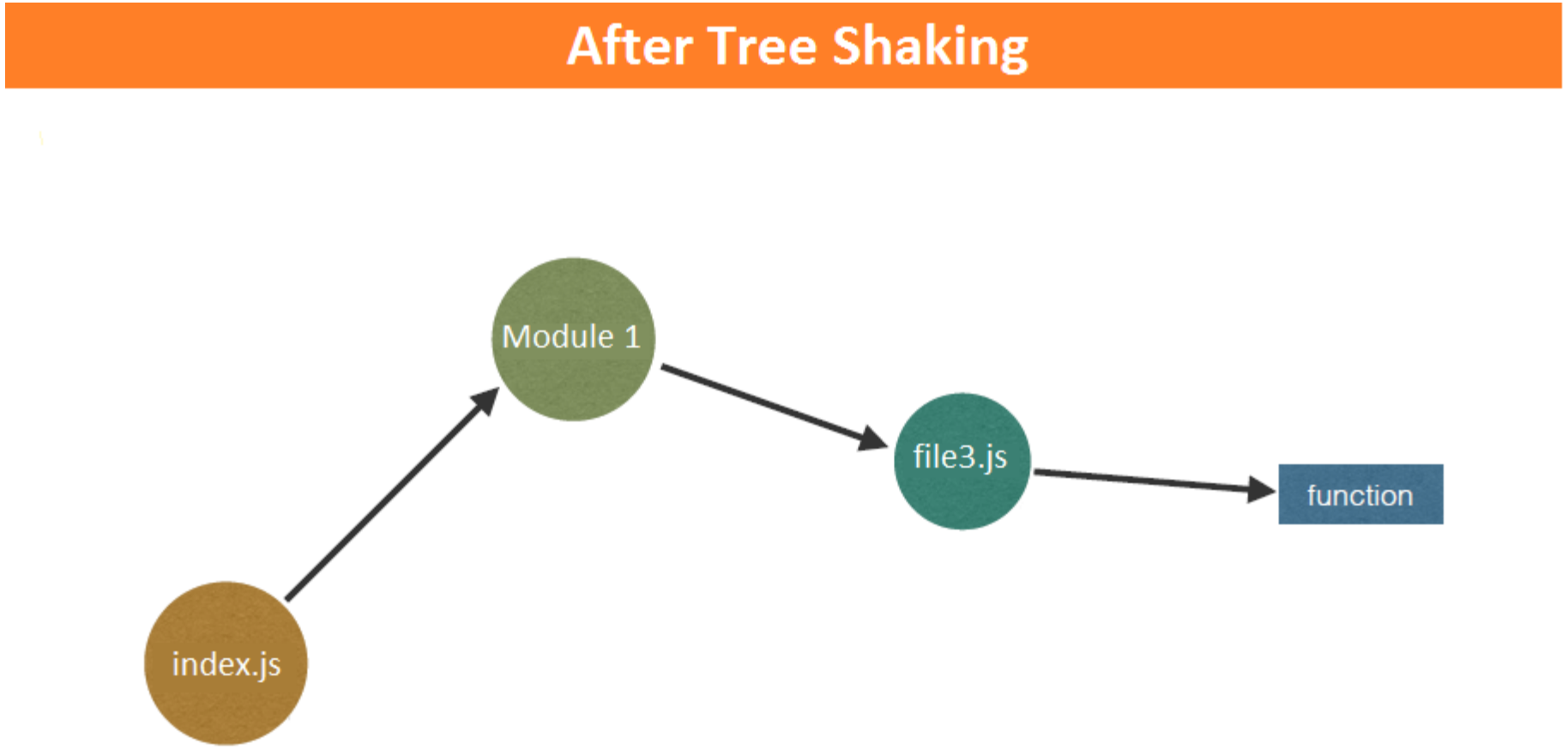
#3: Advantages of Angular Ivy (since V9)

- Angular ViewEngine itself was not tree-shakable
- Default since NG 10, for libs default since NG 12
- AOT per default → You don't need to include the compiler!
- Ivy also does a lot of under the hood optimization
- Tools can easier analyse the code
  - Remove unneeded parts of frameworks
  - Called Tree Shaking
    - Also 3rd party and
    - Even our own libs

#3: Tree Shaking



#3: Tree Shaking



#3: Use Build Optimization – I

Problem:

- *Too large build*
- *Downloading the Angular App takes too much time / ressources*

Identify:

- CSS / JS Files not minimized
- Unused JS code included in the build

#3: Use Build Optimization – II

Solution:

- Use production build
  - ng b(uild) (--c production)

- Set up angular.json correctly

```
"production": {
  "buildOptimizer": true,
  "optimization": true,
  "vendorChunk": true
}
```

DEMO – Build Configuration

#4: Avoid large 3rd party libs / CSS frameworks

- Problem: *Importing large 3rd party libraries that are not treeshakable*
  - *moment*
  - *lodash*
  - *charts*
  - …

- Identify: Source Map Analyzer or Webpack Bundle Analyzer

- Solution: Remove or replace that lib / framework
  - *moment → date-fns*
  - *lodash → lodash-es*
  - …

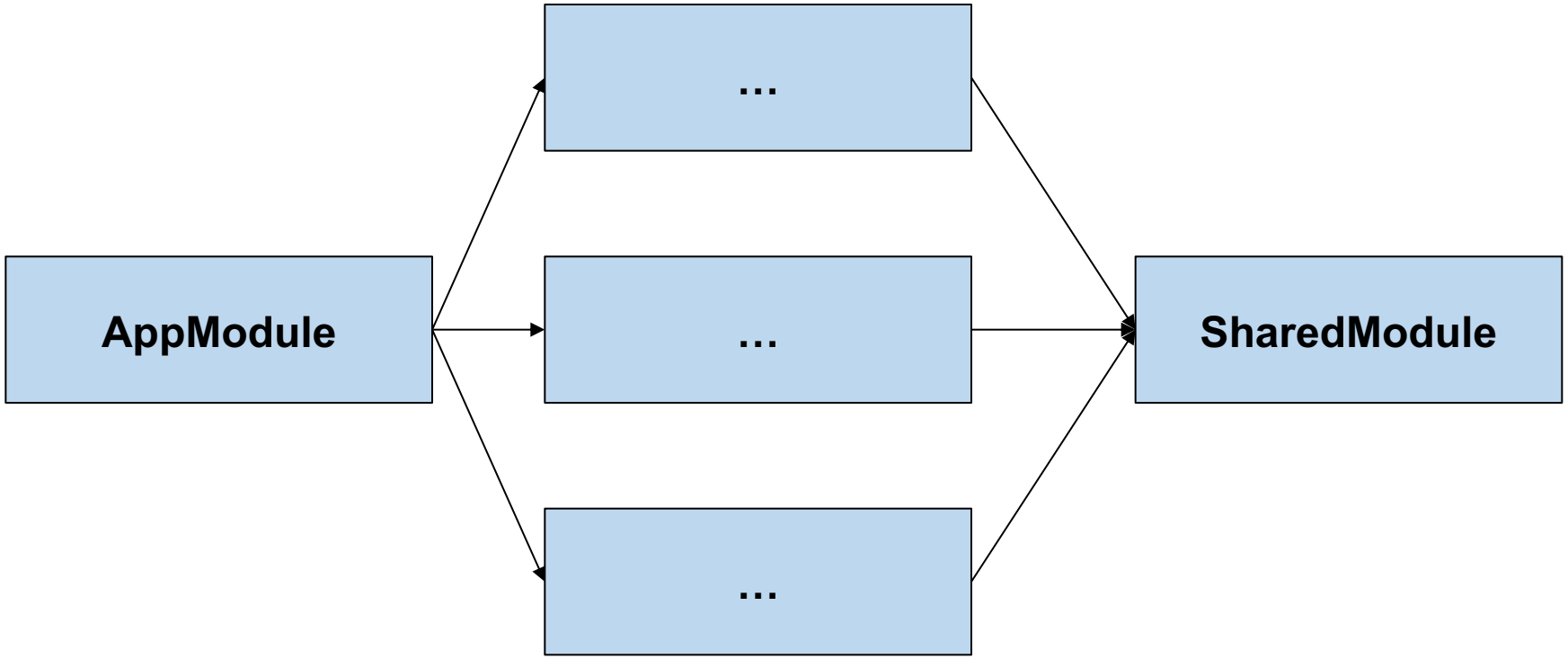
DEMO – Large Libs

Lazy Loading

#5: Angular Lazy Loading

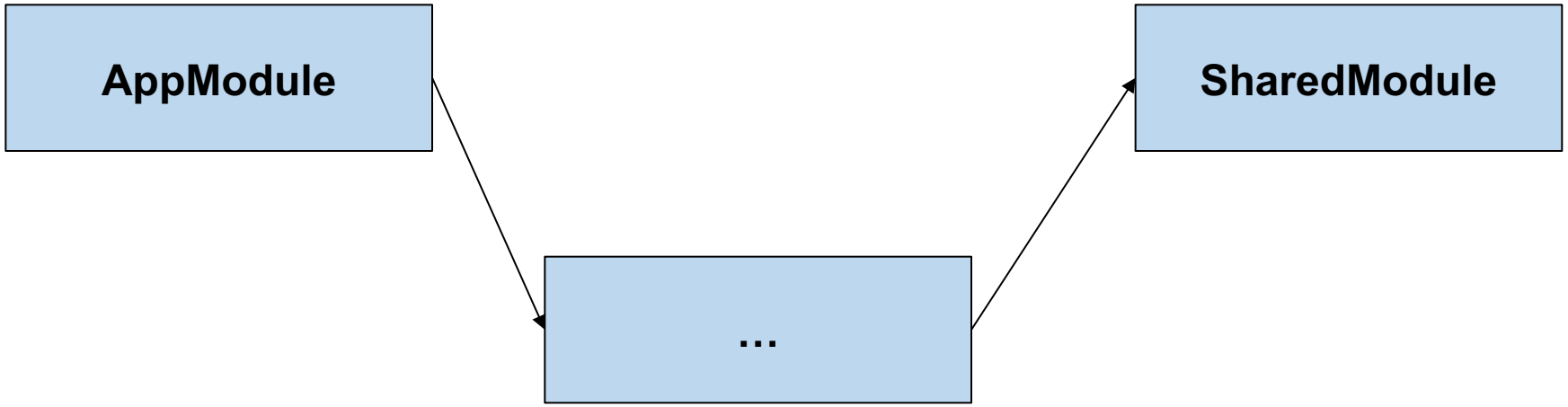
- Lazy Loading means: Load it later, after startup
- Better initial load performance
- But: Delay during execution for loading on demand

#5: Angular Module Structure



Root Module **Feature Modules** **Shared Module**

#5: Angular Lazy Loading



Root Module **Feature Modules** **Shared Module**

#5: AppModule Routes with Lazy Loading

```
const APP_ROUTE_CONFIG: Routes = [
    {
        path: 'home',
        component: HomeComponent
    },
    {
        path: 'flights',
        loadChildren: () => import('./flight-booking/flight-booking.module')
                            .then(m => m.FlightBookingModule)
    } ,
    {
        path: 'charts',
        loadComponent: () => import('./flight-booking/charts.component')
                            .then(c => c.ChartsComponent)
    }
];
```

#5: Routes for "lazy" Module

```
const FLIGHT_ROUTES =    [
    {
        path: 'subroute',
        component: FlightBookingComponent,
        […]
    },
    […]
}
```

flight-booking/ subroute

Triggers Lazy Loading w/ loadChildren

Preloading

#5: Preloading Modules

- Module that might be needed later are loaded after the application started
- When module is needed it is available immediately

#5: Activate Preloading (in AppModule)

```
…
imports: [
    […]
    RouterModule.forRoot(
        ROUTE_CONFIG,
        { preloadingStrategy: PreloadAllModules }
    );
]
…
```

#5: Use Lazy Loading a lot

Problem:

- *Loading to much source (libs / components) at startup*
- *Resulting in a big main bundle (and vendor if used)*

Identify:

- Not using lazy loading throughout the App (source code)
- Webpack Bundle Analyzer or
- Import Graph Visualizer

#5: Use Lazy Loading a lot - but carefully ;-)

Solution:

- Implement lazy loading whereever you can
  - Use lazy loading with the router
    - Modules
    - Components (new since NG15!)
    - Maybe use a CustomPreloadStrategy if App is very big
  - Use dynamic components

- Use Import Graph Visualizer to detect why things land in main bundle

- **But** don't lazyload the initial feature, because it will be delayed ;-)

- **And** don't destroy lazy loading by (eagerly) loading a huge shared module

DEMO – Lazy Loading

#5 Intelligent Preloading with ngx-quicklink

```
…
imports: [
    […]
    QuicklinkModule,
    RouterModule.forRoot(
        ROUTE_CONFIG,
        { preloadingStrategy: QuicklinkStrategy }
    );
]
…
```

https://web.dev/route-preloading-in-angular/

https://www.npmjs.com/package/ngx-quicklink

#5 What about services?

```
…
@Injectable({
  providedIn: 'root'
})
…
```

- When used by 1 lazy loaded module/comp exlusively it will be put into that chunk
- When used by 2 or more lazy loaded modules/comps it will be put into a common chunk
- When used by an eagerly loaded part it will be put into main bundle

#5.5 Lazy Loading without the router

```
…
    @ViewChild('cnt', { read: ViewContainerRef }) vC!: ViewContainerRef;
…
    async ngOnInit() {
        const esm = await import('./lazy/lazy.component');
        const lazyComponentRef = this.vC.createComponent(esm.LazyComponent);
    }
…
```

```
…
    <ng-container #cnt></ng-container>
…
```

#6: Critical Rendering Path / Above the fold

- Problem: *Bad PageSpeed Score that cannot be fixed with #1*

- Identify: Initial load is too slow
  - Using Lighthouse / PageSpeed Insights or
  - WebPageTest

- Solution: Use custom lazy loading of content below the fold
  - Not trivial
  - Has to be implemented manually

Server-side rendering

#7: Server-side rendering (Angular Universal)

- Problem: *After download rendering on the client takes too much time*
  - *Search Engines may not be abled to index the App correctly*

- Identify: After .js files have been loaded js main thread takes too long
  - Search Engines don't index correctly

- Solution: Use Angular Universal
  - Page is rendered on the server and then served to the client

#7.5: Prerender important routes (Universal)

- Problem: *Server response takes to long cos page has to be rendered*
- Identify : Long server response time when using Universal SSR
- Solution: Prerender the important pages on the server
  - Built-in Angular Universal since V11
  - Then serve them rendered to the user

#8: Use a (URL) cache

Alternative Solution:

- Of course you could also use an alternative caching solution
  - E.g. Cloudflare or any other CDN

Lab

Initial Load Performance

Recap

1. **Use web performance best practices**
2. **Use NgOptimizedImage (since v15)**
3. **Use Build Optimization**
4. **Avoid large 3rd party libs / CSS frameworks**
5. **Use Lazy Loading done right**
6. Critical Rendering Path / Above the fold
7. **Server-side rendering & prerender or cache on the server**
8. Use a URL cache

References

- Optimize the bundle size of an Angular application
  - https://www.youtube.com/watch?v=19T3O7XWJkA

- Angular Docs
  - NgOptimizedImage
  - NG Build
  - Lazy-loading feature modules
  - Server-side rendering (SSR) with Angular Universal